Shang Liu

Imperial College London
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EDUCATION

Visiting Ph.D. Student
Department of Management Science and Engineering, Stanford University, Stanford, CA
Oct. 2025 – Jun. 2026
Host: Prof. Jose Blanchet

Ph.D. in Operations and Analytics
Business School, Imperial College London (ICL), London, UK
Sept. 2022 – Jun. 2027 (Expected)
Advisors: Prof. Xiaocheng Li and Prof. Kalyan Talluri

M.Res. in Operations and Analytics (Distinction)
Business School, Imperial College London (ICL), London, UK
Sept. 2021 – Jun. 2022

B.S. in Mathematics and Applied Mathematics
School of Mathematical Sciences, Peking University (PKU), Beijing, China
Sept. 2016 – Jun. 2021

RESEARCH INTERESTS

My research focuses on enhancing the alignment, robustness, and ethical deployment of large language models (LLMs) through interdisciplinary frameworks of human-AI collaboration, game theory, statistical learning, and optimization. I aim to develop both theoretically guaranteed (e.g., convergence analyses, complexity bounds) and empirically valid methodologies. My research interests also lie in the development and theoretical analysis of machine learning models, with a particular focus on the role of uncertainty, including uncertainty quantification, active learning, and human-AI interaction.

PUBLICATIONS

(reverse chronological order; *: equal contribution)

The Critic Is Only a Proxy: Regret Optimization for Reinforcement Learning
Shang Liu*, Yikai Wang*, Guanting Chen, Jose Blanchet, *Working paper; draft available at* OpenReview

Wasserstein Distributionally Robust Regret Optimization for Reinforcement Learning from Human Feedback
Yikai Wang*, **Shang Liu***, Jose Blanchet, arXiv

Incentivizing High-Quality Human Annotations with Golden Questions
Shang Liu, Zhongze Cai, Hanzhao Wang, Zhongyao Ma, Xiaocheng Li, arXiv

How Humans Help LLMs: Assessing and Incentivizing Human Preference Annotators
Shang Liu*, Hanzhao Wang*, Zhongyao Ma, Xiaocheng Li, *under minor revision at* ***Management Science***, arXiv

Reward Modeling with Ordinal Feedback: Wisdom of the Crowd
Shang Liu*, Yu Pan*, Guanting Chen, Xiaocheng Li, ***ICML 2025***, arXiv

Towards Better Understanding of In-Context Learning Ability from In-Context Uncertainty Quantification
Shang Liu[*], Zhongze Cai[*], Guanting Chen, Xiaocheng Li, ***Transactions on Machine Learning Research 2025***, arXiv

Towards Better Statistical Understanding of Watermarking LLMs
Zhongze Cai[*], **Shang Liu**[*], Hanzhao Wang[*], Huaiyang Zhong, Xiaocheng Li, ***Journal of the American Statistical Association 2026***, arXiv

When No-Rejection Learning is Consistent for Regression with Rejection
Xiaocheng Li, **Shang Liu**, Chunlin Sun, Hanzhao Wang, ***AISTATS 2024***, arXiv

Understanding Uncertainty Sampling via Equivalent Loss
Shang Liu, Xiaocheng Li, *reject and resubmit at* ***Journal of Machine Learning Research***, arXiv

Distribution-Free Model-Agnostic Regression Calibration via Nonparametric Methods
Shang Liu[*], Zhongze Cai[*], Xiaocheng Li, ***NeurIPS 2023***, arXiv

Maximum Optimality Margin: A Unified Approach for Contextual Linear Programming and Inverse Linear Programming
Chunlin Sun[*], **Shang Liu**[*], Xiaocheng Li, ***ICML 2023***, arXiv

Non-stationary Bandits with Knapsacks
Shang Liu, Jiashuo Jiang, Xiaocheng Li, ***NeurIPS 2022***, arXiv

Online Bin Packing with Known T
Shang Liu, Xiaocheng Li, *under Major Revision at* ***Mathematics of Operations Research***, arXiv

PRIZES & AWARDS

Dean's List for Academic Excellence at Imperial College Business School, 2021/2022

Outstanding Graduate of Peking University, 2021

Wusi Scholarships at Peking University, 2016/2017, 2017/2018

SERVICE

Reviewer: NeurIPS, ICLR, AISTATS, ICML, AAAI, TMLR, INFORMS IJDS

TEACHING

Teaching Assistant:

Machine Learning, Imperial College London, Spring 2024/2025/2026

Generative AI and Large Language Models, Imperial College London, Spring 2025/2026